# Weather, daylight, snow and bicycle traffic on rue de Terrebonne

## Five count series, August 2024 to November 2026: methods in brief and the main results

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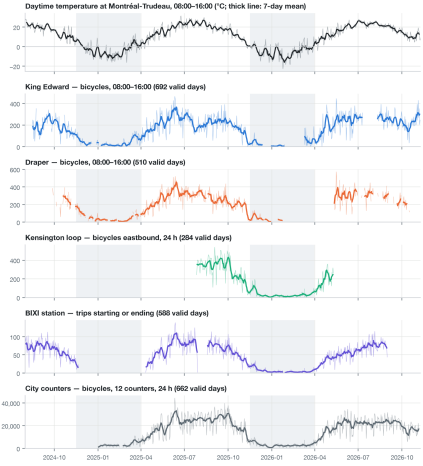
Terrebonne-open-sources

November 10, 2026

# The question and the sources

- How does the number of bicycles counted on rue de Terrebonne vary with temperature, wind chill, rain, snow, daylight and snow loading?
- Five series, each analysed on its own and never compared in level:
  - **King Edward** and **Draper**: citizen-operated Telraam camera sensors, bicycles 08:00–16:00 (691 and 509 valid days from August and September 2024)
  - **Kensington loop**: the borough's counter, eastbound, 24 h (284 days, July 27, 2025 to May 10, 2026; hourly release of June 4, 2026)
  - **BIXI station** Grand Boulevard / de Terrebonne: trips starting or ending, regular season
  - **City counters**: twelve loop counters elsewhere in Montréal, as a reference
- Weather: Environment and Climate Change Canada, Montréal-Trudeau (daily rain, snowfall, snow on the ground; hourly temperature and wind). Snow loading: the City's record for sector CDN-301.

# The record




Each panel has its own scale.

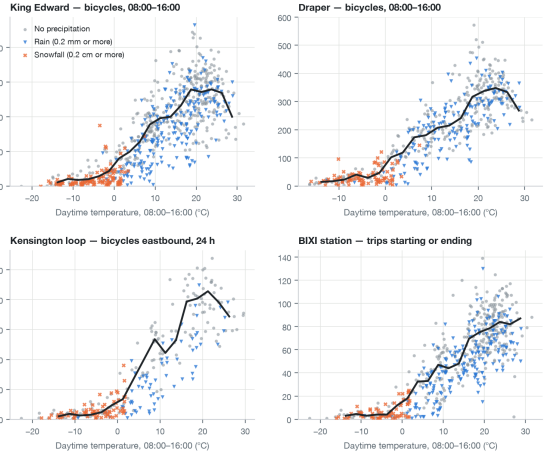
# Method (1): what is counted

- A Telraam sensor counts **only in daylight**: about 9 hours in December, 16 in June. Its daily total moves with the length of the day even if riding does not.
- Every Telraam figure here is the count in a **fixed window, 08:00–16:00**, daylight all year. The evening peak is left out.
- Valid day: at least 8 reported hours and 6 hours of uptime, and all eight window hours with uptime of 0.5 or more. Nothing is interpolated.
- The loop: eastbound channel only (the westbound channel under-counts from about March 1, 2026).
- BIXI: a day without a trip is absent from the file, not a zero; April 15 to November 15 only in the models.

# Method (2): the model

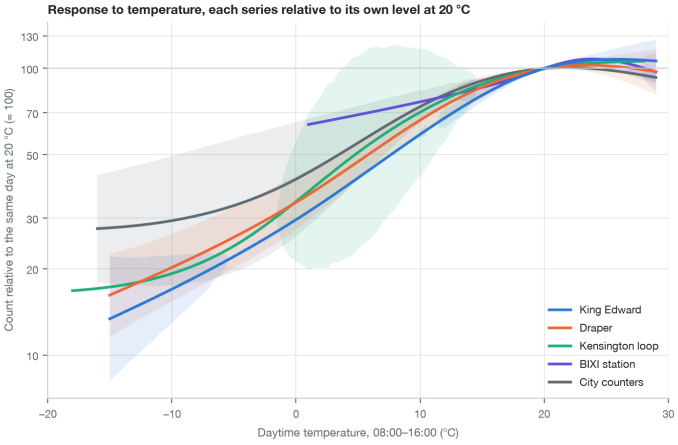
- One **negative binomial regression per series**, log link: every effect is a percentage change, so series can be set side by side without comparing levels.
- Terms: day type, holidays, counting year, time of year (two annual harmonics), daytime temperature, rain and snowfall in classes, snow on the ground, wind, snowfall of the day before.
- Temperature: a spline, and a line with one change of slope whose position is estimated.
- Intervals robust to autocorrelation (Newey–West); moving-block bootstrap for the change of slope.
- Checks: effects re-estimated inside calendar months; each month left out and predicted; wind chill in place of temperature; an hourly model of the loop for darkness.

# Temperature: the raw relation




Each panel has its own scale. Black line: median of the days without precipitation.

# Temperature: the fitted response, relative to 20 °C



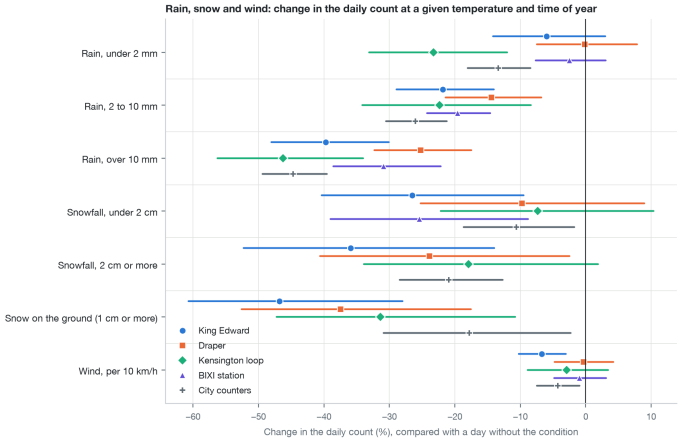

Negative binomial models, one per series, with day type, year, season (two annual harmonics), rain, snowfall, snow on the ground and wind held fixed. Bands: 95 % intervals (autocorrelation-robust). Logarithmic axis: equal vertical steps are equal percentage changes.

Terrebonne series: 5 to 6 % more bicycles per degree up to roughly 20 °C; at 0 °C, 30 to 34 % of the count at 20 °C; at −10 °C, 17 to 20 %.

# Temperature: what the fits say

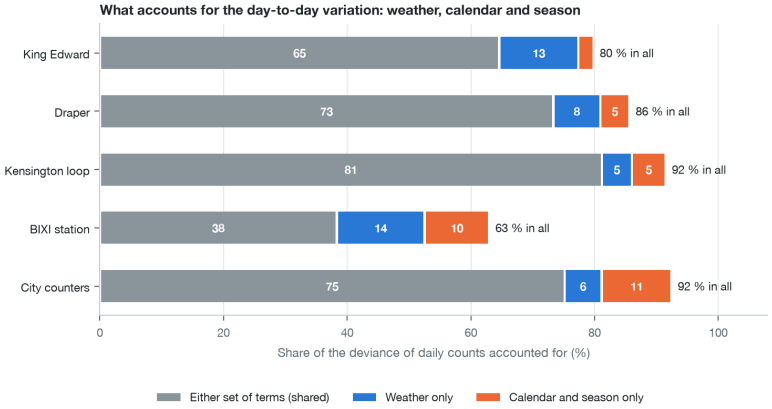
- Below roughly 20 °C, each degree goes with 5 to 6 % more bicycles on the three Terrebonne series; the same inside calendar months.
- Above that the count is flat or falls slightly. The position of the bend is poorly determined (intervals of ten degrees or more): the response bends gradually.
- **No second change of slope in the cold**: below 0 °C the percentage slope is not distinguishable from the slope between 0 and 20 °C.
- The $-1$ and $-5$ °C thresholds of the earlier analyses came from fitting straight lines to the counts themselves.
- Wind chill in place of temperature and wind fits about as well (correlation 0.996); the data cannot separate them.

# Rain, snow and wind at a given temperature and time of year



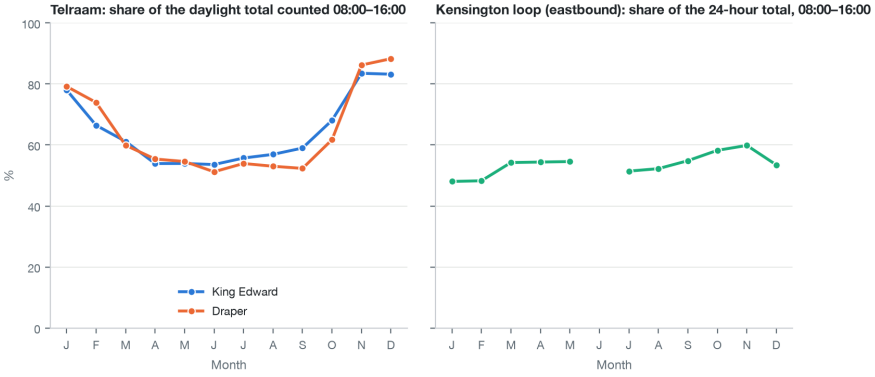

Negative binomial models, one per series; lines: 95 % intervals (autocorrelation-robust). Rain and snowfall are the day's totals at Montréal-Trudeau (24 h). A series has no mark where it has fewer than five such days (BIXI: April 15 to November 15 only). Snowfall of 10 cm or more is merged with 2 to 10 cm.

# Season or weather?




Most of what the models account for can be credited to either: the record cannot fully separate season from weather. Weather effects estimated inside calendar months are the same.

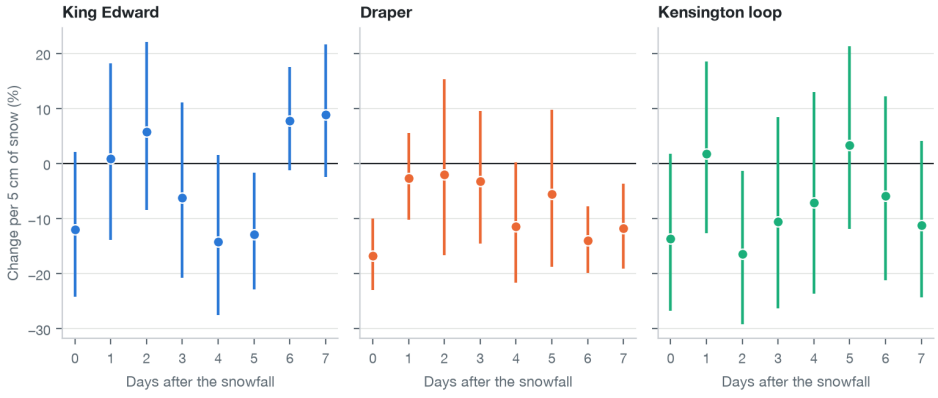
# Day length: what a daylight-only sensor sees




Left: a Telraam sensor counts only in daylight, so its daily total covers 9 hours in December and 16 in June; the fixed window covers the same 8 hours all year.
Right: the loop counts day and night; June has no complete day in the hourly release (July 26, 2025 to May 11, 2026; released June 4, 2026).

On the loop (hourly), the same evening hour carries 27 % (19 to 34) fewer bicycles after sunset than before, at the same temperature and in the same month.

# Snowfall and the days after (winter days)




Negative binomial models on winter days, one per series; lines: 95 % intervals (autocorrelation-robust). Day 0 is the day of the snowfall.

The day of the snowfall shows the clearest reduction; no pattern common to the three series on the following days. Snow on the ground carries part of the after-effect.

# Snow-loading days

| Series | Winter days | Loading days | Change on a loading day (%) |
|---|---|---|---|
| King Edward | 191 | 44 | −4 (−29 to +30) |
| Draper | 154 | 37 | −21 (−48 to +21) |
| Kensington loop | 132 | 36 | +2 (−25 to +39) |

- With temperature, snowfall, snow on the ground, rain, wind and day type in the model, loading days are not distinguishable from other winter days.
- The intervals are wide: they contain no change and a reduction of a quarter or more. The record does not settle the question.

# How well the models predict

| | | Median absolute error (%) | | $R^2$ (log scale) | |
|---|---|---|---|---|---|
| Series | Days | Fitted | Month left out | Fitted | Month left out |
| King Edward | 691 | 22 | 23 | 0.84 | 0.82 |
| Draper | 505 | 19 | 20 | 0.88 | 0.86 |
| Kensington loop | 284 | 20 | 25 | 0.92 | 0.87 |
| BIXI station | 441 | 15 | 17 | 0.59 | 0.50 |
| City counters | 661 | 15 | 18 | 0.94 | 0.93 |

Each calendar month in turn is left out and predicted from the rest.

# Limits

- Telraam figures are 08:00–16:00 counts; never compared in level with the loop.
- Gaps: King Edward has no valid day in February 2026; Draper none from February to April 2026.
- The loop: one site, one winter, one direction.
- Weather at the airport (9 km); precipitation as 24-hour totals; the state of the lane is not observed.
- Season and weather are confounded; the models describe association in the record, not cause.
- Year-to-year terms absorb anything that differs between years besides the weather (gaps, sensor position); see the note.

# Where everything is

- Note, tables, figures (English and French) and scripts: `https://github.com/OpenMobilityData/Terrebonne-open-sources/tree/main/2026-weather-and-bike-traffic`
- Sources: Environment and Climate Change Canada; Telraam (citizen-operated sensors); borough loop counter (access requests 2026-300, released June 4, 2026, and 2026-581, released October 9, 2026); City of Montréal open data (CC BY 4.0); BIXI Montréal open data.